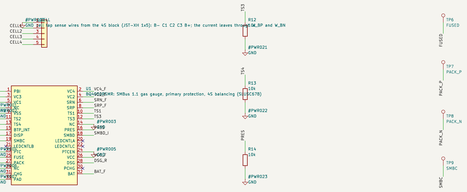
CELL TAPS, GAUGE BQ4050, SENSE AND SUPPLY FILTERS, THERMISTOR



BMS sense wires from the 4S block (JST-XH 1x5): B− C1 C2 C3 B+; the current leaves through W_BK_BP and W_BN

BQ4050RSMR: SMBus 1.1 gas gauge, primary protection, 4S balancing (SLUSD70)

FUSE, PROTECTION FETS, GATE NETWORKS, TERMINAL ESD, PACK LEADS

W_BCELL4
cell block B+ (12 AWG from the block's positive strip)

W_BNBPW9025
cell block B− (12 AWG from the block's negative strip)

F1
25 A mini blade (Keystone 3568 holder): the pack's fuse

CSD17570Q5B 30 V N-FET, charge switch

CSD17570Q5B 30 V N-FET, discharge switch

D1 PACK_P
SMAJ28A

W_PPACK_P
pack lead + (12 AWG to the XT60 on E6 J_BATT pin 2)

W_NPACK_N
pack lead − (12 AWG to the XT60 on E6 J_BATT pin 1)

SMBUS OUT, TEST POINTS, FLAGS

D2
USBLC6-2SC6 ESD clamp on the SMBus pair

J4 SMBus lead to E6 J_SMB (JST-XH 1x4): SMBC SMBD GND PRES

10 kΩ thermistor (JST-PH 1x2): the 103AT in the block

TP1 BTP_INT
TP2 FUSE
TP3 CHG_G
TP4 DSG_G
TP5 SW
TP6 FUSE3
TP7 PACK_P
TP8 PACK_N
TP9 SMBC
TP10 SMBD
#FLG01 PWR_FLAG
#FLG02 PWR_FLAG
#FLG03 PWR_FLAG
#FLG04 PWR_FLAG
#FLG05 PWR_FLAG
#FLG06 PWR_FLAG
#FLG07 PWR_FLAG
#FLG08 PWR_FLAG
#FLG09 PWR_FLAG
#PWR028 GND

Phase P1 schematic (BQ4050 SMBus gauge and protection for the built 4S pack; appendix D2.A2), ge
MeshSat
Sheet 1
File: pcb-a-pack.kicad_sch

Title: MeshSat Field Kit carrier — PCB-P PACK BMS

Size: A0	Date: 2026-04-07	Rev: A
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